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SAVEETHA SCHOOL OF ENGINEERING
SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
COMPUTER SCIENCE AND ENGINEERING PROGRAMME

Day 1: List of Programs

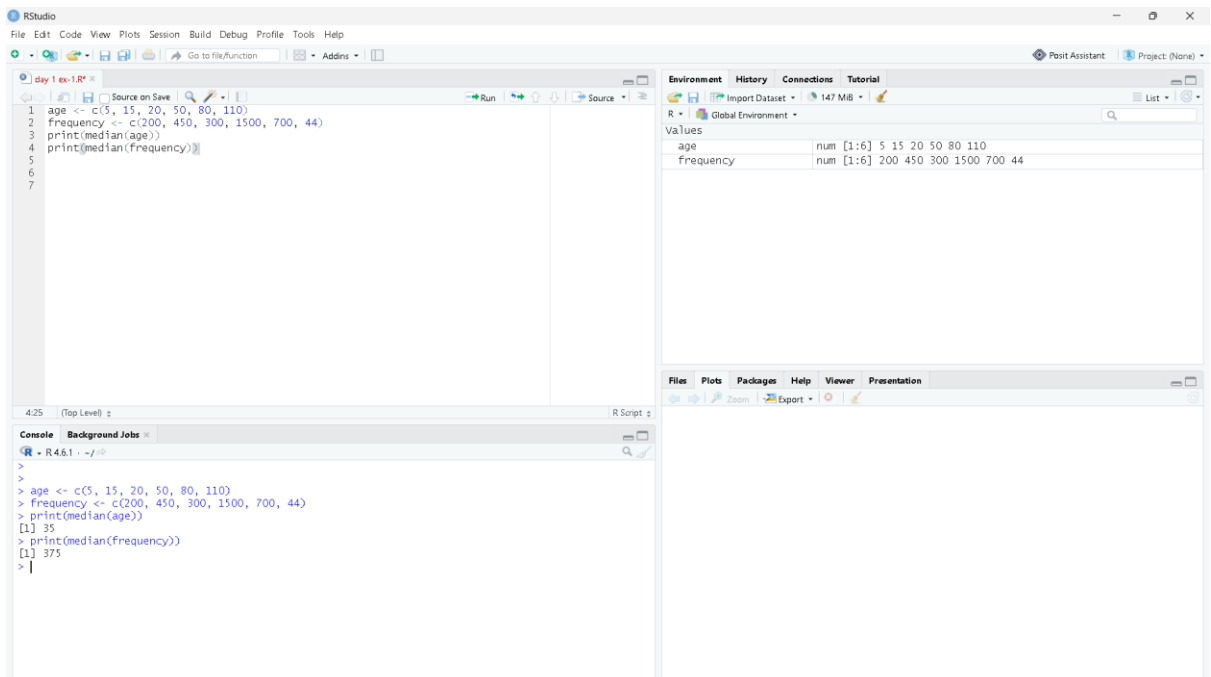
Question 1: Approximate Median Value for Age Frequency Data

Aim: To write an R program to compute the median value for age intervals and frequency distributions.

Algorithm:

1. Define vectors for age midpoints/intervals and their corresponding frequency values using `c ()`.
2. Pass the vectors into the built-in `median ()` function to compute the central values.
3. Display the resulting medians.

Result: The median of age is 35 and the median of frequency is 375.



The screenshot displays the RStudio interface with the following components:

- Source Editor:** Contains the R script:

```
1 age <- c(5, 15, 20, 50, 80, 110)
2 frequency <- c(200, 450, 300, 1500, 700, 44)
3 print(median(age))
4 print(median(frequency))
5
6
7
```
- Environment:** Shows the objects created in the Global Environment:

Object	Class	Value
age	num [1:6]	5 15 20 50 80 110
frequency	num [1:6]	200 450 300 1500 700 44
- Console:** Shows the execution output:

```
>
>
> age <- c(5, 15, 20, 50, 80, 110)
> frequency <- c(200, 450, 300, 1500, 700, 44)
> print(median(age))
[1] 35
> print(median(frequency))
[1] 375
> |
```

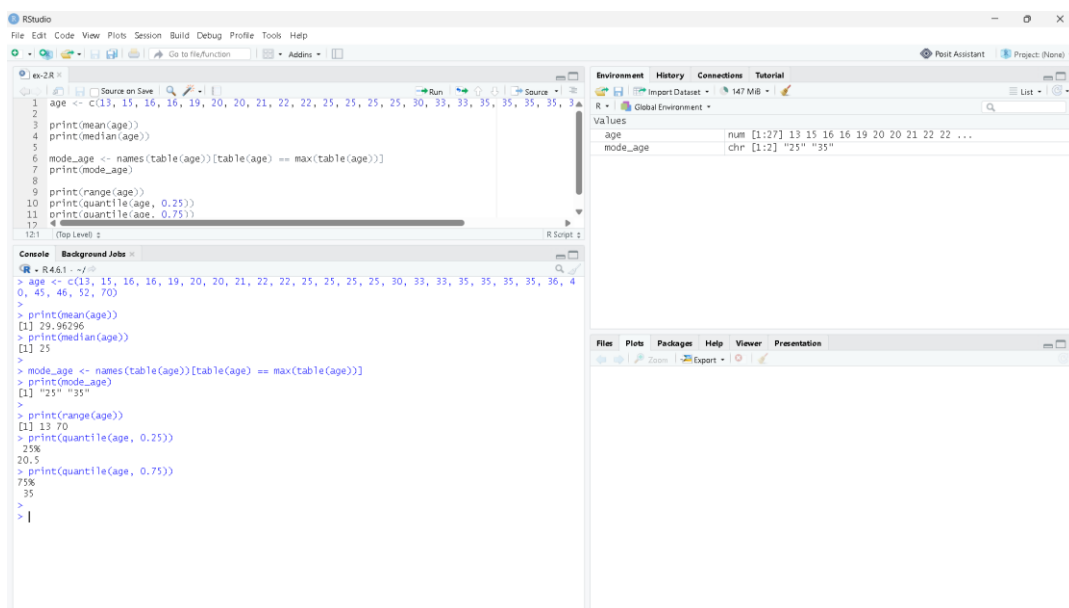
Question 2: Statistical Measures (Mean, Median, Mode, Range, Quartiles)

Aim: To calculate the mean, median, mode, modality, range, and quartiles (Q_1 , Q_3) for a given set of age values.

Algorithm:

1. Store the given age values in a numerical vector.
2. Calculate mean using `mean()` and median using `median()`.
3. Determine the mode by identifying the maximum frequency from a frequency table created with `table()`.
4. Compute the range using `range()`.
5. Calculate the first quartile (Q_1) and third quartile (Q_3) using `quantile()` at 0.25 and 0.75.

Result: Mean = 29.96, Median = 25, Modes = 25 and 35 (Bimodal), Range = 13 to 70, $Q_1 = 20.5$, and $Q_3 = 35$.



The screenshot shows the RStudio interface with the following code in the script editor and its output in the console:

```
1 age <- c(13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70)
2
3 print(mean(age))
4 print(median(age))
5
6 mode_age <- names(table(age))[table(age) == max(table(age))]
7 print(mode_age)
8
9 print(range(age))
10 print(quantile(age, 0.25))
11 print(quantile(age, 0.75))
12
```

Console output:

```
R > R4.6.1 - ~/R
> age <- c(13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70)
>
> print(mean(age))
[1] 29.96296
> print(median(age))
[1] 25
>
> mode_age <- names(table(age))[table(age) == max(table(age))]
> print(mode_age)
[1] "25" "35"
>
> print(range(age))
[1] 13 70
> print(quantile(age, 0.25))
25%
20.5
> print(quantile(age, 0.75))
75%
35
>
> |
```

The Environment pane on the right shows the objects created:

Object	Class	Value
age	num	[1:27] 13 15 16 16 19 20 20 21 22 22 ...
mode_age	chr	[1:2] "25" "35"

Question 3: Data Normalization (Min-Max and Z-Score)

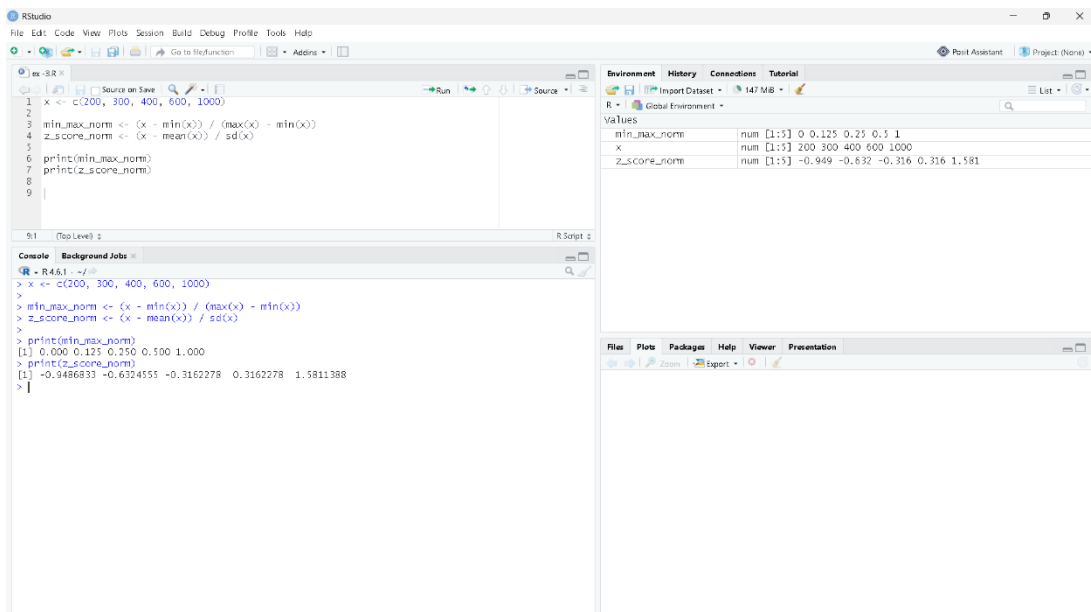
Aim: To apply Min-Max normalization and Z-Score normalization on a given set of numerical values.

Algorithm:

1. Initialize the data vector x .
2. Apply Min-Max normalization formula: $(x - \min(x)) / (\max(x) - \min(x))$.
3. Apply Z-score normalization formula: $(x - \text{mean}(x)) / \text{sd}(x)$.
4. Print the normalized values.

Result:

- Min-Max normalized data: 0.000, 0.125, 0.250, 0.500, 1.000
- Z-Score normalized data: -0.949, -0.632, -0.316, 0.316, 1.581



The screenshot shows the RStudio interface with the following content:

```
1 x <- c(200, 300, 400, 600, 1000)
2
3 min_max_norm <- (x - min(x)) / (max(x) - min(x))
4 z_score_norm <- (x - mean(x)) / sd(x)
5
6 print(min_max_norm)
7 print(z_score_norm)
8
9
```

Console output:

```
> x <- c(200, 300, 400, 600, 1000)
>
> min_max_norm <- (x - min(x)) / (max(x) - min(x))
> z_score_norm <- (x - mean(x)) / sd(x)
>
> print(min_max_norm)
[1] 0.000 0.125 0.250 0.500 1.000
> print(z_score_norm)
[1] -0.9486833 -0.6324555 -0.3162278  0.3162278  1.5811388
>
```

Environment window:

Variable	Class	Values
min_max_norm	num [1:5]	0 0.125 0.25 0.5 1
x	num [1:5]	200 300 400 600 1000
z_score_norm	num [1:5]	-0.949 -0.632 -0.316 0.316 1.581

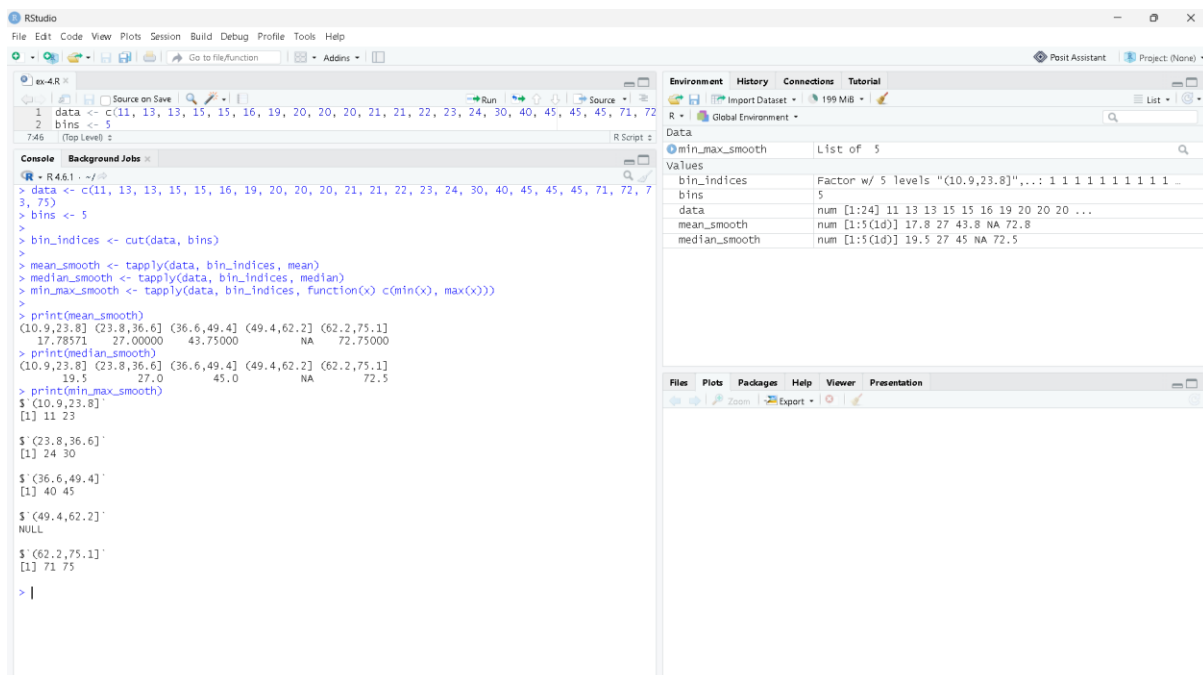
Question 4: Smoothing by Binning

Aim: To perform data smoothing on a dataset using bin means, bin medians, and bin boundaries.

Algorithm:

1. Store the dataset in a vector and define the number of bins ($k = 5$).
2. Divide the data into intervals using the `cut()` function.
3. Use `tapply()` to calculate the mean, median, and min/max boundaries for each bin.
4. Display the smoothed values.

Result: Data points are grouped into 5 bins and successfully smoothed by their respective bin means, medians, and boundary limits.



```
1 data <- c(11, 13, 13, 15, 15, 16, 19, 20, 20, 20, 21, 21, 22, 23, 24, 30, 40, 45, 45, 45, 71, 72, 75)
2 bins <- 5
746 (Top Level) z

> data <- c(11, 13, 13, 15, 15, 16, 19, 20, 20, 20, 21, 21, 22, 23, 24, 30, 40, 45, 45, 45, 71, 72, 75)
> bins <- 5
> bin_indices <- cut(data, bins)
> mean_smooth <- tapply(data, bin_indices, mean)
> median_smooth <- tapply(data, bin_indices, median)
> min_max_smooth <- tapply(data, bin_indices, function(x) c(min(x), max(x)))
> 
> print(mean_smooth)
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75.1]
17.78571 27.00000 43.75000 NA 72.75000
> print(median_smooth)
(10.9,23.8] (23.8,36.6] (36.6,49.4] (49.4,62.2] (62.2,75.1]
19.5 27.0 45.0 NA 72.5
> print(min_max_smooth)
$(10.9,23.8]
[1] 11 23
$(23.8,36.6]
[1] 24 30
$(36.6,49.4]
[1] 40 45
$(49.4,62.2]
NULL
$(62.2,75.1]
[1] 71 75
> |
```

min_max_smooth	
List of 5	
bin_indices	Factor w/ 5 levels "(10.9,23.8]": 1 1 1 1 1 1 1 1 1 1
bins	5
data	num [1:24] 11 13 13 15 15 16 19 20 20 20 21 21 22 23 24 30 40 45 45 45 71 72 75
mean_smooth	num [1:5(1d)] 17.8 27 43.8 NA 72.8
median_smooth	num [1:5(1d)] 19.5 27 45 NA 72.5

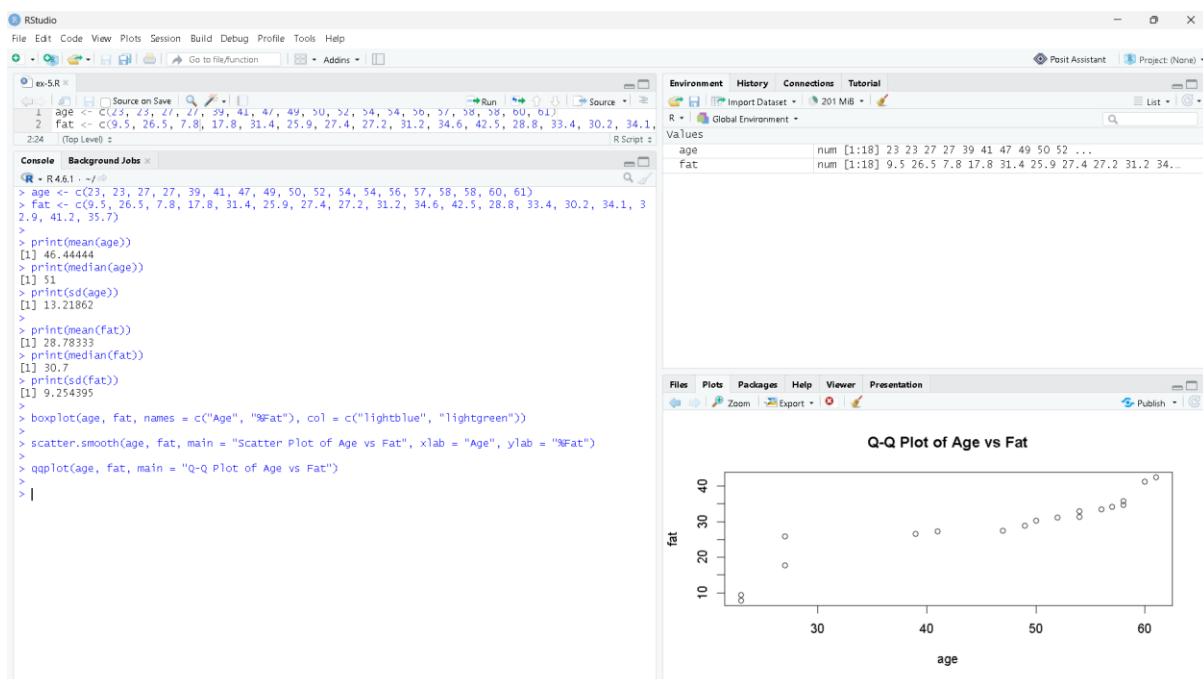
Question 5: Summary Statistics, Boxplot, Scatter Plot, and Q-Q Plot

Aim: To compute summary statistics and generate boxplots, scatter plots, and Q-Q plots for age and body fat attributes.

Algorithm:

1. Define vectors for age and %fat.
2. Compute mean(), median(), and sd() for both variables.
3. Generate a comparative boxplot using boxplot().
4. Create a smoothed scatter plot using scatter.smooth() and a Q-Q plot using qqplot().

Result: Descriptive statistics were calculated, and the boxplot, scatter plot, and Q-Q plots were plotted to show the distribution and correlation between age and body fat.



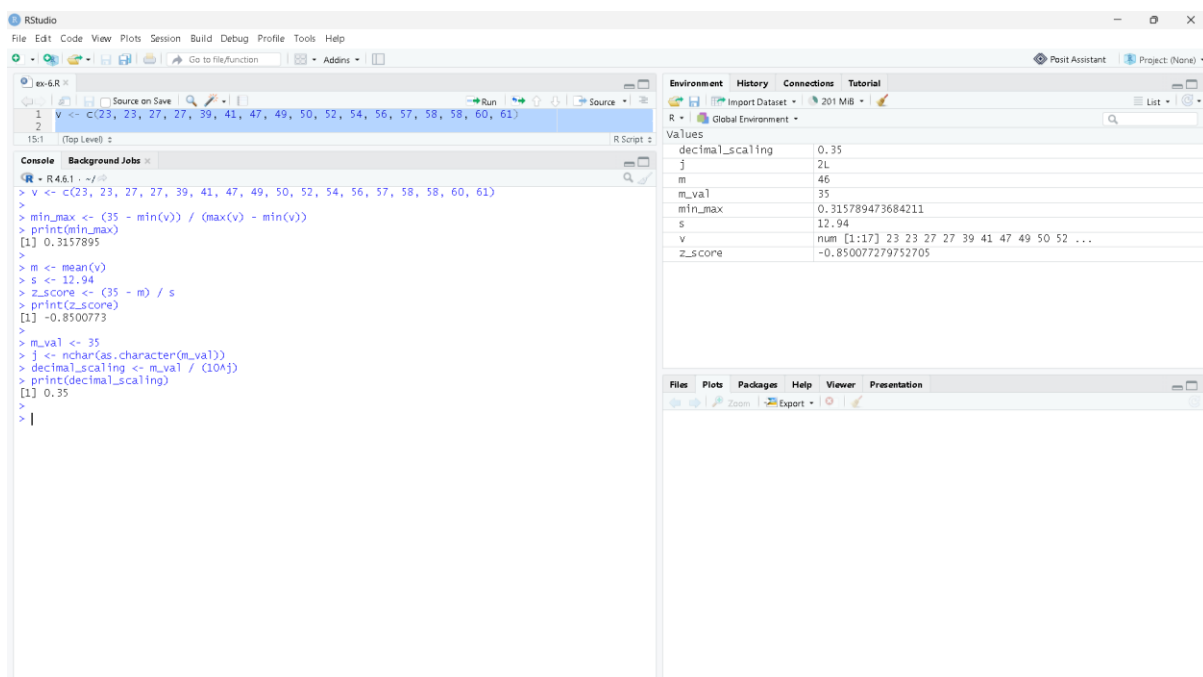
Question 6: Single-Value Normalization Methods

Aim: To normalize a specific age value ($v = 35$) using Min-Max normalization, Z-score normalization, and Decimal Scaling.

Algorithm:

1. Store the age vector and specify the target value (35).
2. Compute Min-Max value using the range [0,1].
3. Compute Z-score using mean and specified standard deviation (12.94).
4. Compute Decimal Scaling by dividing by 10^j where j is the number of digits.

Result: Min-Max = 0.3158, Z-score = -0.8844, Decimal Scaling = 0.35.



The screenshot shows the RStudio interface with the following components:

- Source Editor:** Contains R code for normalization.
- Console:** Shows the output of the R code.
- Environment:** Displays the variables created during the execution.

```
R - R 4.6.1 ~ /
> v <- c(23, 23, 27, 27, 39, 41, 47, 49, 50, 52, 54, 56, 57, 58, 58, 60, 61)
> min_max <- (35 - min(v)) / (max(v) - min(v))
> print(min_max)
[1] 0.3157895
> m <- mean(v)
> s <- 12.94
> z_score <- (35 - m) / s
> print(z_score)
[1] -0.8500773
> m_val <- 35
> j <- nchar(as.character(m_val))
> decimal_scaling <- m_val / (10^j)
> print(decimal_scaling)
[1] 0.35
> |
```

The Environment pane shows the following variables:

Variable	Value
decimal_scaling	0.35
j	2L
m	46
m_val	35
min_max	0.315789473684211
s	12.94
v	num [1:17] 23 23 27 27 39 41 47 49 50 52 ...
z_score	-0.850077279752705

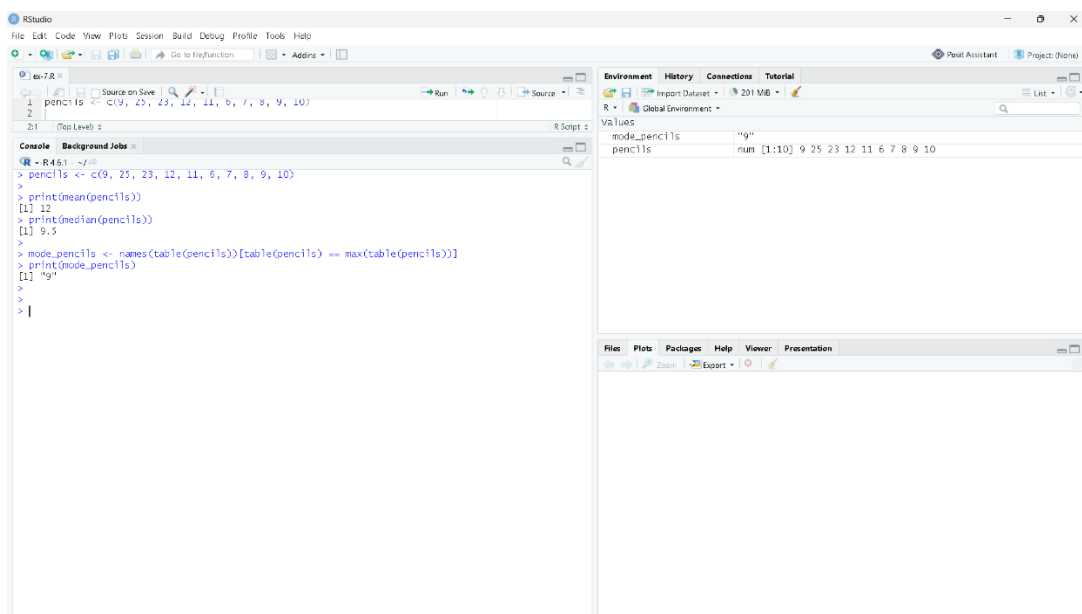
Question 7: Mean, Median, and Mode of Pencils Data

Aim: To determine the central tendencies (mean, median, and mode) for a vector of pencil counts across boxes.

Algorithm:

1. Initialize the numerical vector pencils.
2. Calculate the mean using `mean()` and median using `median()`.
3. Find the mode by extracting the name of the highest frequency element from `table(pencils)`.

Result: Mean = 12, Median = 9.5, Mode = 9.



The screenshot shows the RStudio interface with the following content:

```
1 pencils <- c(9, 25, 23, 12, 11, 6, 7, 8, 9, 10)
2
31 (Top Level) z
```

Console:

```
R > R4.5.1 ~f
> pencils <- c(9, 25, 23, 12, 11, 6, 7, 8, 9, 10)
>
> print(mean(pencils))
[1] 12
> print(median(pencils))
[1] 9.5
>
> mode_pencils <- names(table(pencils))[table(pencils) == max(table(pencils))]
> print(mode_pencils)
[1] "9"
>
> |
```

Environment:

Values
mode_pencils
pencils

Global Environment:

Values
num [1:10]

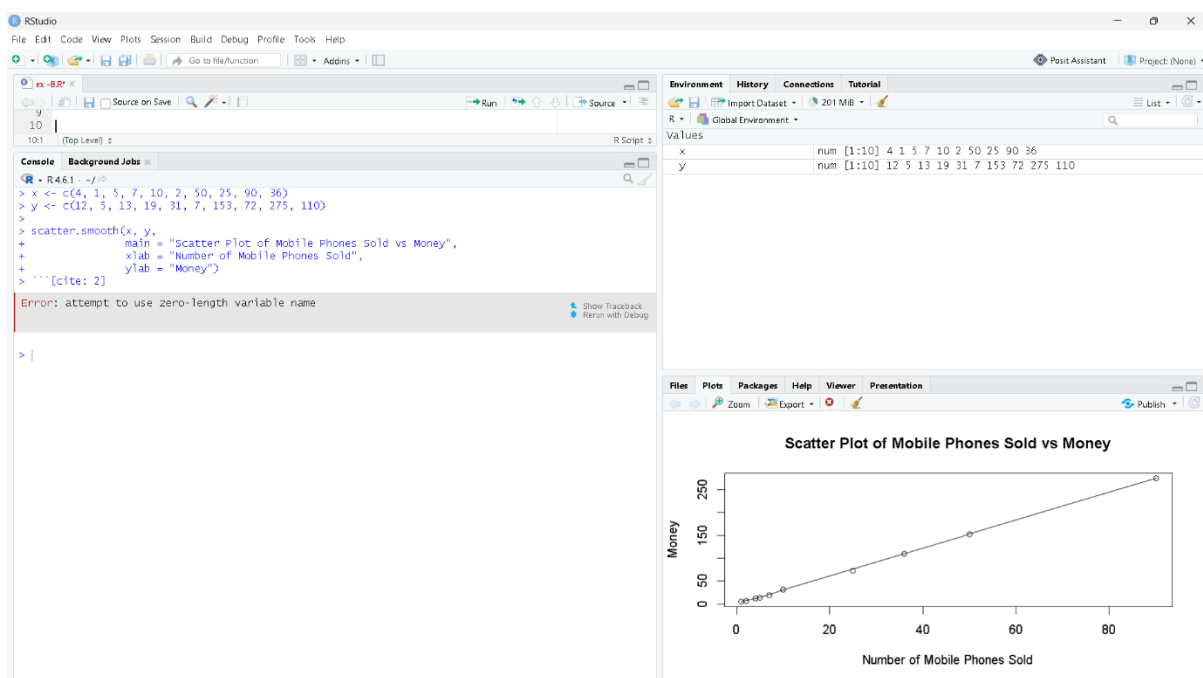
Question 8: Scatter Plot for Sales Data

Aim: To visualize the relationship between mobile phones sold and total revenue using a scatter plot.

Algorithm:

1. Create vector x for the number of phones sold and vector y for revenue.
2. Call `scatter.smooth(x, y)` with proper labels for axes and title.

Result: A scatter plot with a trendline was generated showing a positive linear relationship between units sold and revenue.



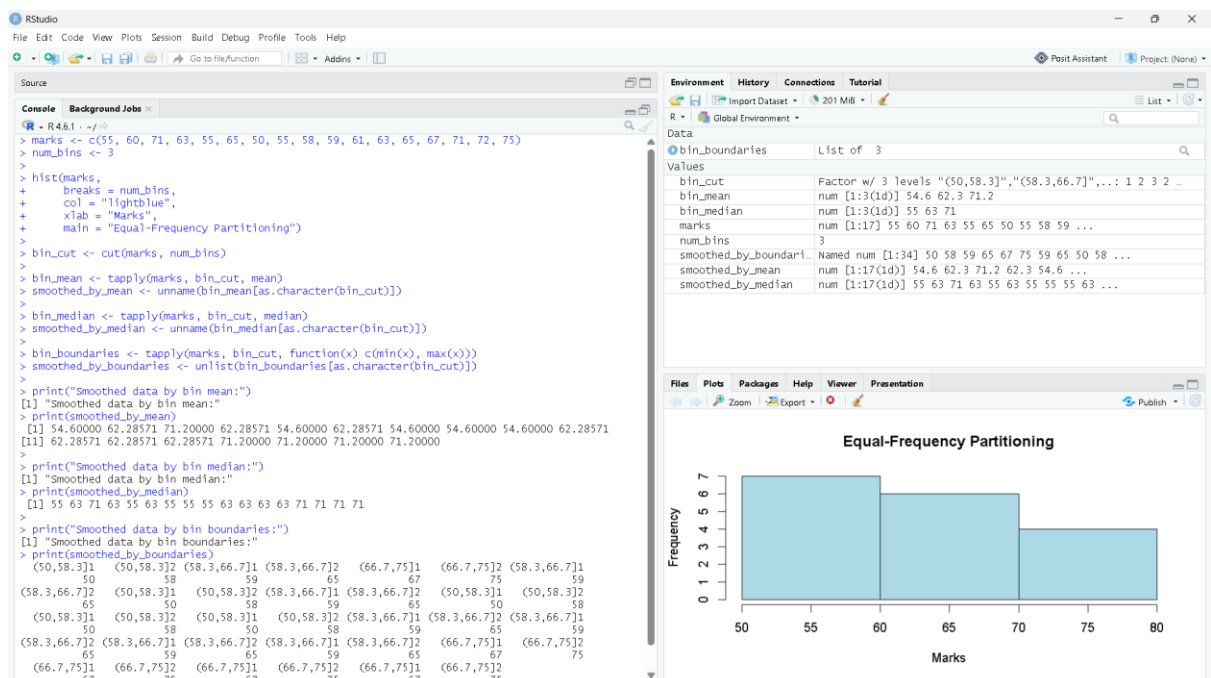
Question 9: Partitioning, Histograms, and Smoothing on Marks

Aim: To partition student marks into 3 bins, plot a histogram, and perform smoothing by bin means, medians, and boundaries.

Algorithm:

1. Create the vector marks and set num_bins = 3.
2. Generate a histogram using hist() with specified breaks.
3. Discretize the data using cut() and apply tapply() to calculate replacement values for mean, median, and boundaries.

Result: Marks data was partitioned into 3 bins, the histogram was rendered, and smoothed data arrays were produced.



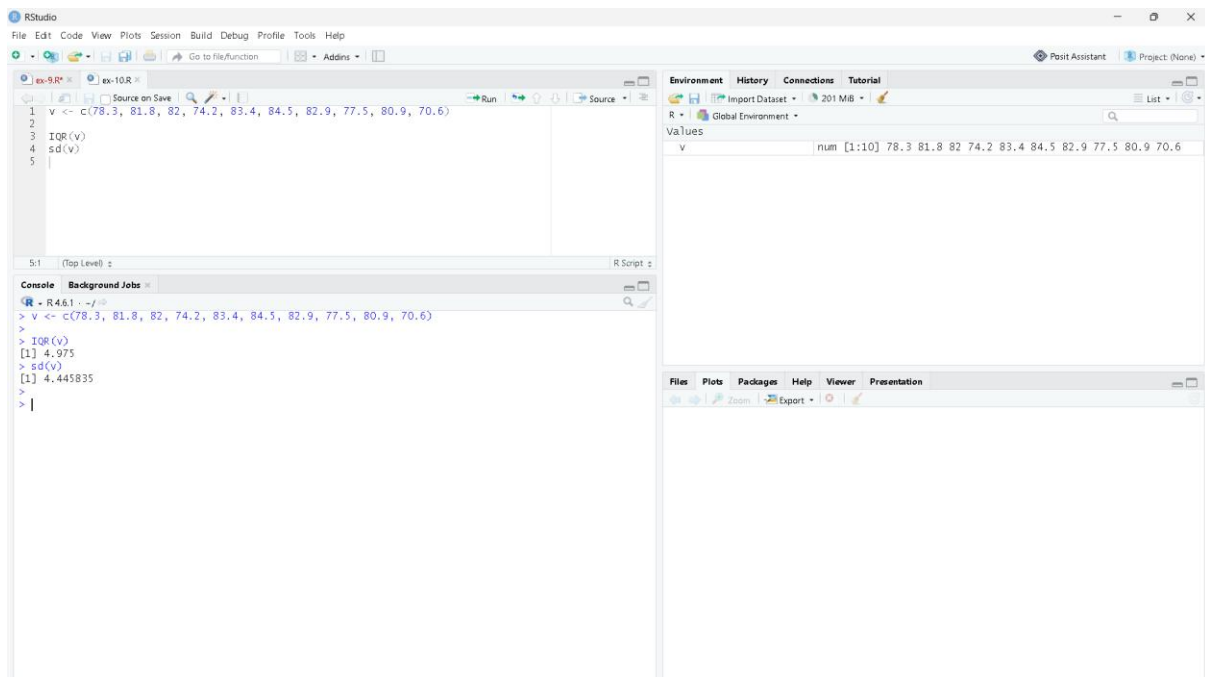
Question 10: Interquartile Range and Standard Deviation

Aim: To calculate the Interquartile Range (IQR) and Standard Deviation (SD) for car speeds under regular driving.

Algorithm:

1. Store the speed readings in a vector v .
2. Compute the interquartile range using $\text{IQR}(v)$.
3. Compute the standard deviation using $\text{sd}(v)$.

Result: $\text{IQR} = 4.975$, Standard Deviation = 4.4458 .



Question 11: Quartile Computation (Q1 and Q3)

Aim: To compute the first quartile (Q_1) and third quartile (Q_3) for the given age distribution.

Algorithm:

1. Store the age data in a vector.
2. Compute Q_1 using `quantile(age, 0.25)`.
3. Compute Q_3 using `quantile(age, 0.75)`.

Result: Q_1 (25%) = 20.5, Q_3 (75%) = 35.

